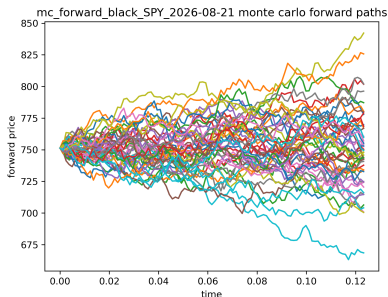


SABR model validation

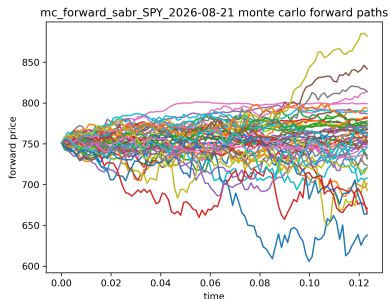
Noorain Noorani and Ruojun Wang

July 10, 2026

Forward path Prices



(a) Black model forward paths.



(b) SABR model forward paths.

Figure: Sample Monte Carlo forward price paths generated under the Black and SABR models. For clarity, only the first 50 simulated paths are shown, while all simulated paths are retained for pricing and hedging calculations.

Pricing and Hedging

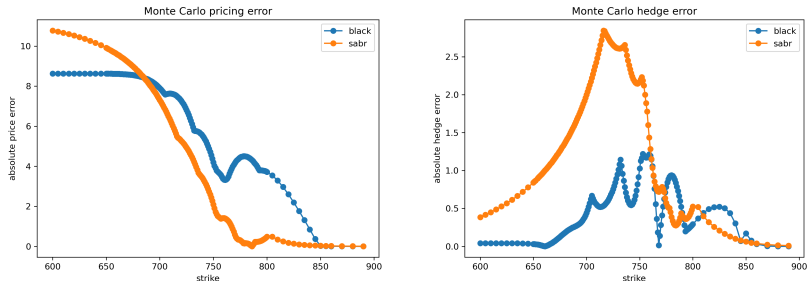


Figure: Comparison of the Monte Carlo pricing error and dynamic hedging error for the Black and SABR models across different strike prices. The left panel shows the absolute pricing error between the Monte Carlo estimate and the corresponding closed form option price, while the right panel shows the absolute hedging error obtained from the dynamic delta hedging simulation.

Distribution of hedging values

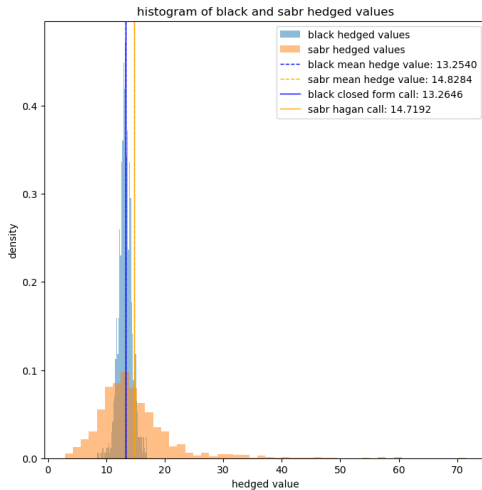
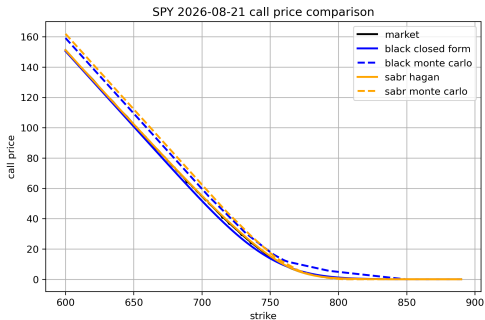


Figure: Strike $K = 751.0$

Pricing errors against market



black closed form rmse : 1.937651

black monte carlo rmse : 4.963974

sabr hagan rmse : 0.339667

sabr monte carlo rmse : 6.201467